

Machine Technical Manual: High-Speed Thermosonic Ball-Wedge Wire Bonder

Domain Intelligence & Anomaly Diagnostics Specification for Model WB-001

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SYNTHETIC PROTOTYPE TECHNICAL MANUAL — VECTORAI SYSTEM SPECIFICATION

This document is artificially generated specifically for the VectorAI predictive maintenance demonstration, knowledge base retrieval (RAG), formula-based RUL calculation, and automated diagnostics.

The specifications, thresholds, sensor values, service-life hours, maintenance intervals, and operating parameters are synthetic and must NOT be used for real-world physical industrial equipment operation or maintenance.

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1. Document Information & Governance

This technical document provides authoritative machine specifications for **High-Speed Thermosonic Ball-Wedge Wire Bonder** (Identifier: WB-001, Manual ID: VAI-MAN-WB-001). It establishes the single source of truth for the VectorAI platform, defining telemetry sensor schemas, threshold limits, maintenance intervals, and deterministic degradation parameters.

Governance Principle: All diagnostic algorithms, RUL models, and threshold engines in VectorAI must strictly adhere to the limits and parameters defined herein. No machine learning regressions or stochastic models are permitted in the primary RUL calculation pipeline.

2. Machine Overview & Manufacturing Process

Process Stage: Wire Interconnect Assembly

Ultra-fast thermosonic ball-wedge wire bonder designed to create microscopic electrical interconnects between semiconductor die pads and leadframe fingers/substrate traces using gold (Au) or copper (Cu) wire via high-frequency ultrasonic energy, controlled mechanical force, and stage heating.

Detailed Process Flow:

The wire bonding sequence operates at up to 25 wires per second. An Electronic Flame-Off (EFO) spark creates a molten Free Air Ball (FAB) at the tip of the wire protruding from a ceramic capillary. The bond head drives the ball into the die pad, applying 138 kHz ultrasonic resonance to form a metallurgical intermetallic bond (First Bond). The capillary then forms a precision wire loop and presses the wire onto the leadframe finger, creating a crescent/wedge bond (Second Bond) before wire clamps tear the tail.

Key Machine Subsystems:

- 138 kHz Piezoelectric Ultrasonic Transducer & Horn Stack
- Precision Ceramic Capillary Tooling & Z-Axis Voice Coil
- Electronic Flame-Off (EFO) Spark Generator & Electrode
- High-Precision Workholder Heated Clamp & Vacuum Stage
- Wire Feed Spool & Micro-Air Tensioner Delivery System
- High-Speed Look-Down Vision Alignment & Wire Loop Profiling Camera

3. Major Components & Degradation Characteristics

Component	Primary Function	Important Parameters	Degradation Indicators
Ultrasonic Transducer & Horn Stack	Converts high-frequency electrical drive signals into mechanical ultrasonic resonance to scrub and weld gold/copper wire to metal pads.	Ultrasonic vibration magnitude (mm/s), resonance frequency (kHz), transducer temperature (°C), impedance (Ω).	Vibration harmonic drift (>0.75 mm/s), piezo stack heating (>58°C), impedance increase, unbonded wires.
Ceramic Capillary Tool	Guides the ultra-fine bonding wire (18-25 μm diameter), shapes the ball bond, and presses the stitch bond.	Tip outer diameter (μm), hole diameter (μm), face angle, touchdown impact force (gf).	Capillary tip wear/erosion, gold buildup in chamfer, touchdown force elevation (>28 gf), non-stick on pad.
Electronic Flame-Off (EFO) Wand	Discharges a high-voltage, low-current micro-arc to melt the wire tip into a symmetrical Free Air Ball.	Spark voltage (V), spark current (mA), spark duration (μs), electrode gap distance (mm).	Spark voltage spikes (>2450 V), asymmetrical ball formation, electrode carbonization/oxidation.
Leadframe Workholder & Clamp	Rigidly clamps leadframe against heated pedestal block to eliminate micro-motion during ultrasonic scrubbing.	Clamping force (N), clamp insert flatness, heater stage temperature (°C).	Clamping force degradation (>85 N or <45 N), leadframe bouncing, poor stitch adhesion, clamp wear.
Wire Feed Clamp & Air Tensioner	Maintains constant light tension on bonding wire spool and executes precision tail breaking.	Air tension pressure (kPa), clamp opening/closing response time (ms), wire drag friction.	Wire sagging, loop height variance, wire drag friction increase, tail length inconsistency.

4. Sensor Configuration & Telemetry Mapping

The following sensor schema represents the continuous telemetry parameters streamed by the machine controller into the VectorAI Edge Collector. Direction indicates whether increasing or decreasing values signify degradation.

Sensor ID	Sensor Name	Unit	Normal Range	Warning Range	Critical Range	Direction
vibration_ultrasonic	Ultrasonic Vibration	mm/s	0.3 – 0.5	0.5 – 0.75	0.75 – 2.0	HIGHER IS WORSE
load_clamp	Clamp Clamping Force	N	55.0 – 75.0	75.0 – 85.0	85.0 – 120.0	HIGHER IS WORSE
temperature_transducer	Transducer Temp	°C	42.0 – 50.0	50.0 – 58.0	58.0 – 80.0	HIGHER IS WORSE
ultrasonic_power	Ultrasonic Power Output	W	1.8 – 2.8	2.8 – 3.6	3.6 – 6.0	HIGHER IS WORSE
efo_spark_voltage	EFO Spark Voltage	V	1800.0 – 2200.0	2200.0 – 2450.0	2450.0 – 3000.0	HIGHER IS WORSE
capillary_touchdown_force	Touchdown Impact Force	gf	18.0 – 24.0	24.0 – 28.0	28.0 – 50.0	HIGHER IS WORSE

5. Sensor Threshold Definitions & Operational Boundaries

Sensor / Metric	Normal Operational Band	Warning Threshold Band	Critical Fault Band
Ultrasonic Vibration vibration_ultrasonic	0.3 to 0.5 mm/s Resonant harmonic energy properly transmitted through horn and capillary.	0.5 to 0.75 mm/s Acoustic impedance mismatch or loosened capillary locking screw.	0.75 to 2.0 mm/s Severe transducer piezo cracking or uncontrolled mechanical oscillation.
Clamp Clamping Force load_clamp	55.0 to 75.0 N Firm clamping holding leadframe flat against heater block without substrate damage.	75.0 to 85.0 N Excessive clamping force causing leadframe indentation or clamp actuator fatigue.	85.0 to 120.0 N Excessive force causing leadframe deformation or lead pin crushing.
Transducer Temp temperature_transducer	42.0 to 50.0 °C Nominal thermal equilibrium under continuous 20 W/sec bonding load.	50.0 to 58.0 °C Piezoelectric internal dielectric loss and self-heating.	58.0 to 80.0 °C Thermal depolarization hazard for PZT crystals; imminent transducer failure.
Ultrasonic Power Output ultrasonic_power	1.8 to 2.8 W Standard energy required to establish Au-Al or Cu-Al intermetallic bond.	2.8 to 3.6 W Generator boosting power to overcome bonding interface contamination or capillary wear.	3.6 to 6.0 W Excessive power delivery risking pad metal cratering and silicon fracture.
EFO Spark Voltage efo_spark_voltage	1800.0 to 2200.0 V Clean, repeatable arc breakdown forming spherical FAB.	2200.0 to 2450.0 V Electrode gap widening or tip oxidation increasing ionization potential.	2450.0 to 3000.0 V EFO arc failure or unstable spark causing golf-club shaped deformed balls.
Touchdown Impact Force capillary_touchdown_force	18.0 to 24.0 gf Gentle touchdown preserving delicate underlying low-k dielectric under die pads.	24.0 to 28.0 gf Voice-coil velocity dampening drift causing hard touchdown impacts.	28.0 to 50.0 gf Severe impact force causing micro-crater fractures beneath bond pads.

6. Standard Operating Conditions & Environmental Limits

Ambient Temperature:	20.0 - 23.0 °C (Cleanroom ISO Class 5 / Class 100)
Relative Humidity:	35% - 50% RH non-condensing (strictly controlled for Cu wire oxidation prevention)
Normal Operating Temperature:	180.0 - 220.0 °C (Workholder heater for Au wire; 150-180°C for Cu wire)
Operating Pressure:	0.45 - 0.55 MPa (CDA & 95% N2 / 5% H2 forming gas for Cu wire)
Operating Speed / Velocity:	18 - 25 wires per second
Standard Cycle Time:	0.04 - 0.06 seconds per bonded wire
Production Schedule:	20 - 24 hours/day continuous
Recommended Operating Conditions:	Forming gas flow 0.5 L/min during EFO fire; capillary clean cycle every 100,000 bonds.
Max Continuous Operation:	100 operating hours before capillary inspection/replacement

7. Maintenance Schedule & Expected Service Life

Component	Interval (Hours)	Expected Life	Required Maintenance Action & Procedure
Ceramic Capillary Tool	100 hrs	250 hrs	Inspect tip wear with optical scope, clean tip with ceramic polishing sheet, replace when worn. <i>Procedure:</i> Torque capillary locking screw to precisely 0.35 Nm using preset torque wrench. Verify vertical orientation.
Ultrasonic Transducer Horn	1000 hrs	5000 hrs	Perform electrical impedance analyzer scan; clean horn mounting interface and check resonance curve. <i>Procedure:</i> Verify resonance peak is at 138.0 ± 0.5 kHz with series resistance < 25 Ω.
EFO Wand & Electrode	200 hrs	2000 hrs	Clean electrode tip with fine abrasive pad, inspect gap setting to wire (target 0.8 mm), check high-voltage cable. <i>Procedure:</i> Measure spark gap with optical reticle. Fire 20 sample sparks; verify ball symmetry under 100x scope.
Leadframe Clamp Insert	300 hrs	3000 hrs	Inspect clamp edge for gold dust contamination and mechanical burrs; verify clamping force with load cell. <i>Procedure:</i> Place load cell wafer under clamp; calibrate clamp motor current to deliver exactly 65 N.
Wire Feed Spool & Tensioner	150 hrs	4000 hrs	Clean sapphire wire guides, replace air tensioner filter, verify zero wire snagging along feed path. <i>Procedure:</i> Inspect wire under microscope for scratches. Verify tension gauge reads 3.0 ± 0.5 gf.

8. Degradation Indicators & Physics of Failure

Each parameter below reflects physical mechanical or electrical wear in machine subsystems. The VectorAI anomaly engine monitors these indicators to calculate Remaining Useful Life (RUL).

Parameter	Normal Baseline	Degraded Condition	Critical Limit	Underlying Physics / Mechanism
Ultrasonic Vibration	Vibration amplitude 0.30 - 0.50 mm/s with sharp 138 kHz resonance.	Vibration drifts into 0.50 - 0.75 mm/s due to capillary screw loosening or horn impedance rise.	Vibration exceeds 0.75 mm/s; transducer resonance breaks down into chaotic noise.	Indicates physical micro-cracking in piezoelectric crystal stack or mechanical loosening of the capillary clamping screw.
Transducer Temperature	Operates steadily between 42 and 50 °C.	Temperature climbs to 50 - 58 °C as internal electrical dissipation increases.	Temperature exceeds 58 °C; approaching Curie temperature depolarization limit.	Indicates internal dielectric breakdown and heat dissipation from micro-fractured piezo ceramic disks.
Ultrasonic Power Output	Power stable between 1.8 and 2.8 W.	Power creeps up to 2.8 - 3.6 W to overcome bonding interface contamination or capillary tip wear.	Power exceeds 3.6 W; high risk of pad metal lifting and silicon micro-cracking.	Compensates for loss of acoustic scrubbing efficiency caused by eroded capillary tip chamfer.
Clamp Clamping Force	Force maintained at 55 - 75 N.	Force creeps into 75 - 85 N due to actuator motor friction.	Force exceeds 85 N or drops below 40 N; leads move during bonding.	Indicates wear in the clamping toggle linkage, pneumatic piston friction, or clamp spring fatigue.
Touchdown Impact Force	Gentle touchdown force between 18 and 24 gf.	Impact force increases to 24 - 28 gf.	Impact force exceeds 28 gf; dangerous shock loads delivered to active die circuitry.	Indicates loss of voice-coil velocity feedback damping or friction in Z-axis linear air-bearing slides.

9. Deterministic Formula-Based RUL Model Configuration

Base Useful Life (BUL): 5000 operating hours

Mathematical Formula: VectorAI computes RUL deterministically without neural networks or stochastic regression. For each parameter *i*:

- For HIGHER_IS_WORSE: $d_i = (currentValue - healthyLimit) / (criticalLimit - healthyLimit)$
- For LOWER_IS_WORSE: $d_i = (healthyLimit - currentValue) / (healthyLimit - criticalLimit)$
- Clamp: $d_i = \max(0.0, \min(1.0, d_i))$
- Overall Degradation: $D_{total} = \sum (w_i \times d_i)$ [where $\sum w_i = 1.00$]
- Remaining Useful Life: $RUL = \max(0, BaseUsefulLife \times (1.0 - D_{total}))$

Parameter	Sensor ID	Unit	Weight (w_i)	Healthy Limit	Critical Limit	Direction
Ultrasonic Vibration	vibration_ultrasonic	mm/s	0.30	0.5	1.1	HIGHER IS WORSE

Transducer Temperature	temperature_transducer	°C	0.20	50.0	68.0	HIGHER IS WORSE
Ultrasonic Power	ultrasonic_power	W	0.20	2.8	4.0	HIGHER IS WORSE
Clamp Clamping Force	load_clamp	N	0.15	75.0	95.0	HIGHER IS WORSE
Touchdown Force	capillary_touchdown_force	gf	0.15	24.0	32.0	HIGHER IS WORSE
TOTAL WEIGHT SUM	-	-	1.00 (100%)	-	-	VERIFIED OK

10. Troubleshooting & Failure Symptoms Table

The following troubleshooting matrix provides rapid diagnostic mappings between observable anomalies and root causes.

Symptom & ID	Severity	Related Sensors	Possible Causes	Recommended Corrective Action
Ultrasonic Transducer Resonance Drift / Vibration Spike SYM-WB-01	Critical	vibration_ultrasonic, temperature_transducer	- Capillary fastening screw torque relaxation (< 0.35 Nm) - Piezoelectric crystal disk micro-cracking or electrode delamination - Transducer horn mounting bracket misalignment	Retorque capillary screw to 0.35 Nm. If vibration persists, execute impedance sweep and replace transducer stack.
Non-Stick on Pad (NSOP) / First Bond Lift-Off SYM-WB-02	Critical	ultrasonic_power, vibration_ultrasonic, capillary_touchdown_force	- Ceramic capillary tip chamfer wear or gold clogging - Substrate surface organic contamination or pad aluminum oxidation - Insufficient ultrasonic power delivery	Replace ceramic capillary tool, execute capillary tip alignment, verify substrate heater temperature (200°C).
Electronic Flame-Off (EFO) Spark Failure / Tail Open SYM-WB-03	High	efo_spark_voltage	- EFO electrode tip carbonization or oxidation buildup - Electrode spark gap out of specification (> 1.2 mm) - Forming gas flow deficiency causing copper wire oxidation	Clean electrode tip with abrasive pad, readjust gap to 0.8 mm, verify 0.5 L/min forming gas flow.
Second Bond (Stitch) Peeling / Non-Stick on Lead (NSOL) SYM-WB-04	High	load_clamp, ultrasonic_power	- Leadframe clamp insert wear or insufficient clamping force allowing substrate bounce - Heater stage pedestal surface particulate buildup - Capillary outer radius wear	Inspect leadframe clamping force (target 65 N), clean heater pedestal surface, replace capillary.
Die Pad Metal Cratering / Silicon Damage SYM-WB-05	Critical	capillary_touchdown_force, ultrasonic_power	- Z-axis touchdown impact force exceeding 28 gf - Ultrasonic power burst applied prior to complete touchdown seating - Excessive ultrasonic power amplitude (> 3.6 W)	Calibrate Z-axis touchdown damping and velocity profile, reduce ultrasonic power by 15%, inspect pad under acoustic scope.

11. Comprehensive Diagnostic Knowledge Base (Failure Scenarios)

This section contains 13 detailed synthetic failure scenarios used as the initial primary diagnostic knowledge base by the VectorAI Machine Agent. When an anomaly is detected, the agent queries this manual first before fallback to RAG.

Scenario 1: Transducer Piezo Stack Micro-Crack Breakdown (SCEN-WB-001)	Severity: CRITICAL
Telemetry Sensor Pattern:	vibration_ultrasonic rises from 0.44 to 0.88 mm/s; temperature_transducer hits 56.5°C
Possible Root Causes:	Repetitive mechanical shock fatigue on piezo crystal discs High-voltage drive amplifier transient spike Capillary overtightened beyond 0.50 Nm
Recommended Corrective Action:	Replace ultrasonic transducer horn assembly. Calibrate generator resonant frequency to 138 kHz.
Verification & Recovery Steps:	Run impedance analyzer scan (target 138.0 kHz, < 25 Ω) > Execute 100 test bonds with wire pull test (> 8 gf)
Scenario 2: Capillary Tip Chamfer Erosion and Gold Buildup (SCEN-WB-002)	Severity: HIGH
Telemetry Sensor Pattern:	ultrasonic_power creeps from 2.2W to 3.4W; NSOP rate climbs to 1.5%
Possible Root Causes:	Abrasive friction from gold/copper wire after 150,000 bond cycles Gold buildup in capillary wire hole causing feed drag Capillary touched off-center on hard leadframe edge
Recommended Corrective Action:	Replace capillary with new ceramic tool. Torque to 0.35 Nm. Re-teach optical bond locations.
Verification & Recovery Steps:	Inspect capillary tip under microscope > Perform wire pull and ball shear test (> 25 gf shear)
Scenario 3: EFO Electrode Oxidation Causing Misshaped Free Air Ball (SCEN-WB-003)	Severity: HIGH
Telemetry Sensor Pattern:	efo_spark_voltage rises to 2520 V; ball diameter variance increases
Possible Root Causes:	Forming gas shield tube misaligned Electrode tip oxidized from ambient air exposure Spark duration timing drift
Recommended Corrective Action:	Polish electrode tip, readjust spark gap to 0.8 mm, verify 95% N2 / 5% H2 gas shield.
Verification & Recovery Steps:	Fire 50 test balls onto leadframe > Verify FAB diameter is 35 ± 1.5 μm under vision inspection
Scenario 4: Leadframe Clamp Wear Causing Substrate Resonance Bounce (SCEN-WB-004)	Severity: HIGH
Telemetry Sensor Pattern:	load_clamp reads 84.0 N; ultrasonic_power elevated; NSOL stitch peels
Possible Root Causes:	Clamp insert contact pads worn and grooved by leadframe strips Clamp air cylinder seal friction Heater block vacuum channels blocked by solder residue
Recommended Corrective Action:	Replace leadframe clamp insert. Clean heater pedestal vacuum channels with brass scraper and IPA.
Verification & Recovery Steps:	Test clamping force with calibrated load cell (target 65 N) > Check stitch bond pull strength (> 5 gf)

Scenario 5: Z-Axis Voice Coil Damping Loss & Pad Cratering Hazard (SCEN-WB-005)	Severity: CRITICAL
Telemetry Sensor Pattern:	capillary_touchdown_force surges to 31.0 gf; bond force alarm
Possible Root Causes:	Z-axis voice-coil velocity feedback circuit gain drift Linear air slide bearing particulate contamination Touchdown sensor optical flag bent
Recommended Corrective Action:	Clean Z-axis air bearing slides, recalibrate voice coil velocity damping profile, verify 20 gf touchdown.
Verification & Recovery Steps:	Perform touchdown test on piezoelectric sensor block > Acoustic microscopy check for cratering
Scenario 6: Wire Spool Tensioner Air Damper Instability (SCEN-WB-006)	Severity: MEDIUM
Telemetry Sensor Pattern:	Wire loop height variance exceeds $\pm 15 \mu\text{m}$; cycle time erratic
Possible Root Causes:	Air tensioner glass tube dirty with gold dust flakes Tension air regulator pressure fluctuation Wire spool drag brake pad worn
Recommended Corrective Action:	Clean glass tensioner tube with alcohol swab. Inspect spool brake spring.
Verification & Recovery Steps:	Measure loop profile on 30 sample wires with vision system (target height $120 \pm 3 \mu\text{m}$)
Scenario 7: Substrate Heater Block Zone 2 Thermal Degradation (SCEN-WB-007)	Severity: HIGH
Telemetry Sensor Pattern:	Bond temperature fluctuates between 165°C and 198°C ; heater duty cycle alert
Possible Root Causes:	Heater cartridge internal resistance aging Thermocouple loose in heater pocket Thermal paste dried and calcified
Recommended Corrective Action:	Replace heater cartridge element, apply high-temperature thermal paste, reseal thermocouple.
Verification & Recovery Steps:	Measure temperature profile across heater block with surface pyrometer (uniformity $\pm 2^{\circ}\text{C}$)
Scenario 8: Wire Clamping Solenoid Response Latency (SCEN-WB-008)	Severity: MEDIUM
Telemetry Sensor Pattern:	Tail length fluctuates ($50 \mu\text{m}$ to $250 \mu\text{m}$); occasional tail-less open alarm
Possible Root Causes:	Wire clamp solenoid pivot jewel bearing dirt Clamp driving pulse voltage drop Clamp ruby inserts contaminated with wire drawing lubricant
Recommended Corrective Action:	Clean wire clamp ruby jaws with optical solvent wipe. Calibrate clamp open/close timing.
Verification & Recovery Steps:	Verify tail length consistency (target $100 \pm 10 \mu\text{m}$) over 500 wires
Scenario 9: Vision System Pattern Recognition Offset Drift (SCEN-WB-009)	Severity: MEDIUM
Telemetry Sensor Pattern:	Die pad alignment search time increases by 40 ms; occasional false PR rejects
Possible Root Causes:	Look-down camera objective lens coated with leadframe flux vapor LED coaxial illumination brightness decay Thermal drift of optical mounting bracket
Recommended Corrective Action:	Clean camera front glass with lens tissue. Execute illumination auto-calibration and optics zeroing.
Verification & Recovery Steps:	Run PR test on 50 reference dies > Verify location accuracy $< 0.5 \mu\text{m}$

Scenario 10: Capillary Clamping Screw Torque Relaxation (SCEN-WB-010)	Severity: HIGH
Telemetry Sensor Pattern:	vibration_ultrasonic exhibits erratic noise spikes (0.35 to 0.72 mm/s)
Possible Root Causes:	Thermal expansion cycling of aluminum horn threads Capillary screw thread stripped or worn Improper torque wrench used during last tool change
Recommended Corrective Action:	Replace clamping screw with new titanium screw. Torque to exactly 0.35 Nm.
Verification & Recovery Steps:	Verify vibration stability during continuous 1000-wire test run
Scenario 11: Forming Gas Flow Interruption During Cu Bonding (SCEN-WB-011)	Severity: CRITICAL
Telemetry Sensor Pattern:	efo_spark_voltage rises to 2600 V; copper wire ball heavily oxidized (dark red/purple)
Possible Root Causes:	Forming gas mass flow controller valve stuck closed Gas line cylinder pressure depleted Gas delivery nozzle tip clogged with spatter
Recommended Corrective Action:	Inspect gas supply line, clear nozzle orifice, verify 0.5 L/min flow rate with inline rotameter.
Verification & Recovery Steps:	Inspect copper FAB under microscope (must be bright shiny copper sphere)
Scenario 12: Second Bond Stitch Cutter Blade Chipping (SCEN-WB-012)	Severity: HIGH
Telemetry Sensor Pattern:	Wire tail fails to break cleanly; wire dragging across package leads
Possible Root Causes:	Tear clamp stroke distance insufficient Capillary outer radius chipped on lead edge Leadframe metal burrs catching wire
Recommended Corrective Action:	Replace capillary tool. Check tear clamp motion profile.
Verification & Recovery Steps:	Verify clean tail separation without wire necking or sagging
Scenario 13: Heater Pedestal Vacuum Suction Loss (SCEN-WB-013)	Severity: MEDIUM
Telemetry Sensor Pattern:	load_clamp at 72 N; leadframe vibration detected during scrub
Possible Root Causes:	Vacuum generator venturi clogged with cleanroom dust Heater pedestal vacuum manifold seal O-ring degraded Leadframe strip bottom surface burrs preventing vacuum seal
Recommended Corrective Action:	Clean vacuum venturi nozzle, replace high-temp silicone O-ring, check leadframe incoming quality.
Verification & Recovery Steps:	Verify vacuum level < -70 kPa under clamped leadframe

12. Standard Operating & Maintenance Procedures (SOP)

SOP-01: Shift Start Operational Inspection & Tool Touch-Off

1. Verify Clean Dry Air (CDA) supply pressure is within 0.55 - 0.65 MPa and dew point is < -40°C.
2. Ensure cleanroom ambient temperature is stabilized within specified operating window (20 - 24°C).
3. Run automated sensor zero-calibration and optical target homing sequence.
4. Process 2 calibration test pieces; inspect output dimensions and telemetry stability on VectorAI Dashboard.

SOP-02: Tool Replacement & Calibration Protocol

1. Lock out and tag out (LOTO) machine controller before entering mechanical enclosure.
2. Use calibrated preset torque wrench to fasten tool fasteners to exact manufacturer specification.
3. Execute optical non-contact tool alignment and measure runout / contact resistance.
4. Log maintenance record in VectorAI with work order ID, technician ID, and timestamp.

SOP-03: Anomaly Escalation Protocol

When a sensor crosses from Warning into Critical threshold band:

1. VectorAI generates automated Severity: High/Critical Incident.
2. System automatically references Section 11 Diagnostic Knowledge for recommended immediate actions.
3. If Remaining Useful Life (RUL) falls below 48 hours, dynamic production rerouting is triggered to alternate healthy lines.

13. Document Revision History

Revision	Release Date	Author / Organization	Description of Changes
v1.0	2026-08-26	VectorAI Knowledge Engineering	Initial synthetic technical specification and diagnostic manual release.
v0.9-RC	2026-08-15	VectorAI Systems Lab	Draft sensor thresholds, RUL weight configuration, and failure scenarios.

14. VectorAI RAG Integration & Knowledge Indexing

This document is formatted with distinct semantic headings, tables, and structured diagnostic scenarios to facilitate high-precision chunking and vector embedding in the VectorAI RAG pipeline.

- **Chunk Association:** Every indexed chunk retains metadata: machineId, manualId, section.
- **Scoped Retrieval:** Diagnostics for machine WB-001 prioritize chunks where machineId == 'WB-001'.